The story of a sunspot at microwaves

How radiosynth computes thermal gyroresonance emission
from the magnetic field of a model

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radiosynth 2.2.0 — *From MHD to MHz* — https://github.com/soumitrahazra/radiosynth

Point a radio telescope at a sunspot group and tune it to a few gigahertz. Over the spots you find a bright source with a brightness temperature of *two to three million kelvin*, strongly circularly polarised. The hot, thin corona above the spots cannot do this by free–free emission alone: for the active-region model used below, free–free gives at most 0.45 MK. Something else is shining. That something is the magnetic field itself — or rather, the electrons spinning in it. This note tells how, and how radiosynth computes it, with only the equations that are really needed.

1. Electrons that spin

The gyrofrequency

In a magnetic field an electron does not move in a straight line: it spirals around the field line. It goes around at a fixed rate, the *gyrofrequency*,

$$\nu_B = \frac{eB}{2\pi m_e} = 2.8\,\mathrm{MHz} \times \frac{B}{1\,\mathrm{G}}. \tag{1}$$

Above a sunspot, where B is hundreds to thousands of gauss, ν_B is in the gigahertz range — exactly where microwave telescopes observe. A spinning charge radiates, and it radiates (and absorbs) at its spinning frequency and at its multiples, the *harmonics* $s\nu_B$, $s = 1, 2, 3, \ldots$

In one sentence: electrons spiral in the field at $\nu_B = 2.8\,\mathrm{MHz}$ per gauss and talk to waves at $s\nu_B$.

2. Thin glowing shells

Where the resonance happens

An observer at a fixed frequency ν only "sees" electrons that resonate with it, where

$$\nu = s\,\nu_B \qquad \Longleftrightarrow \qquad B = B_s = \frac{\nu}{s \times 2.8\,\mathrm{MHz/G}}. \tag{2}$$

Since the field gets weaker with height above a spot, each harmonic picks out one surface of constant $|B|$ — a thin glowing shell, the *gyroresonance layer*. At 5 GHz the shells are at

harmonic s	1	2	3	4
field B_s at 5 GHz	1786 G	893 G	595 G	447 G

with the $s = 4$ shell highest and the $s = 1$ shell deepest. The shells are thin because the resonance is sharp: it is only blurred by the thermal motion of the electrons along the field, so a shell is about 1–2 % of the magnetic scale length $L_B = |B/(\mathrm{d}B/\mathrm{d}\ell)|$ thick — typically about a hundred kilometres.

In one sentence: at each frequency, the emission comes from a few thin shells of constant field strength, $B = \nu/(2.8\,\mathrm{MHz} \times s)$.

3. How opaque is a shell?

The one formula that matters

The optical depth of a shell, how strongly it blocks (and therefore emits) light, is

$$\boxed{\tau_s = \frac{\pi e^2}{m_e c}\,\frac{n_e\,L_B}{\nu}\,\frac{s^2}{s!}\left(\frac{s^2\beta_T^2\sin^2\theta}{2}\right)^{s-1} F(\theta)}, \qquad \beta_T^2 = \frac{k_{\rm B}T_e}{m_e c^2}. \tag{3}$$

It looks heavy, but it tells a simple story:

- $n_e L_B$: more electrons and a thicker shell mean more absorption;
- $\beta_T^2 \approx 3\times10^{-4}$ at 2 MK is a small number, and it is raised to the power $s-1$: **each higher harmonic is several hundred times weaker**. Only the lowest harmonics matter;
- $\sin^2\theta$, where θ is the angle between the field and the line of sight: a shell seen **along** the field is almost transparent, a shell seen **across** it is most opaque;
- $F(\theta)$ depends on how the wave is polarised (Chapter 4).

For typical active-region values ($n_e = 10^9\,\mathrm{cm}^{-3}$, $L_B = 10^9$ cm, $T_e = 2$ MK, 5 GHz, $\theta = 60°$) the code finds

$$\tau_2 \approx 7000, \qquad \tau_3 \approx 12, \qquad \tau_4 \approx 0.03.$$

So the second and third harmonic shells are walls, and the fourth is nearly transparent.

In one sentence: only the shells $s = 2$ and 3 are opaque; each higher harmonic is hundreds of times weaker, and shells seen along the field are much thinner.

4. Two ways to wiggle

The x-mode and the o-mode

In a magnetised plasma a radio wave can only travel in two ways, the two *modes*. In the **extraordinary (x) mode** the electric field rotates in the same sense as the spinning electrons; in the **ordinary (o) mode** it rotates the other way (or oscillates along the field). The x-mode "dances in step" with the electrons and is absorbed far more strongly. This is the factor F in Eq. (3): along the field $F_x = 2$ and $F_o = 0$; at 60° the x-mode shell is typically 20–50 times more opaque than the o-mode one (Fig. 1). Seen along the field, the two modes are circularly polarised, one right-handed and one left-handed.

In one sentence: the x-mode, which rotates with the electrons, sees opaque shells; the o-mode sees much thinner ones.

5. An opaque shell is a thermometer

Kirchhoff's law

The electrons are thermal, so each mode glows exactly as much as it absorbs. Each mode carries half of the thermal radiation,

$$S = \frac{\nu^2 k_{\rm B} T_e}{c^2}, \tag{4}$$

and an opaque shell shows, in that mode, the brightness temperature $T_b = T_e$ of the gas *at the shell*. That is why a sunspot at microwaves is as hot as the corona above it: the telescope

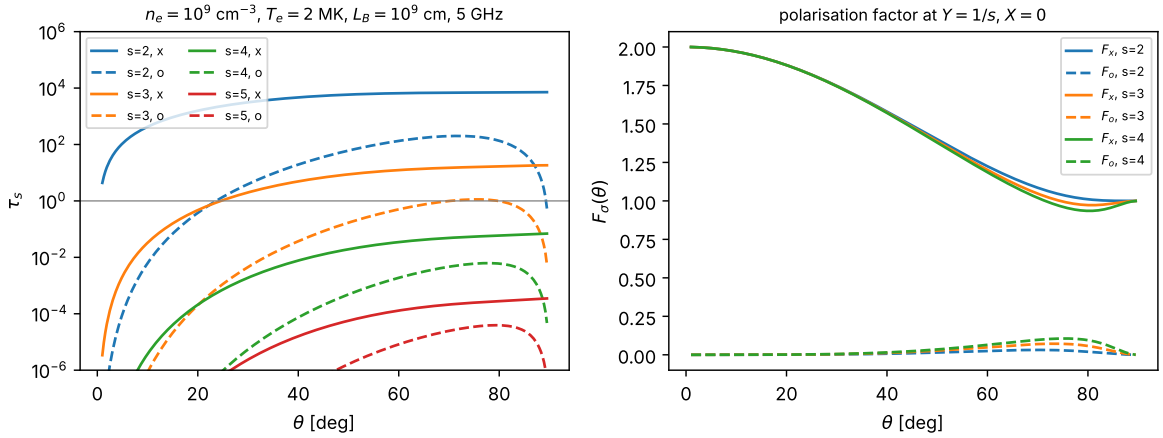


Figure 1: Left: optical depth of the shells $s = 2$–5 against the angle θ between the field and the line of sight (solid: x-mode, dashed: o-mode; same values as in Chapter 3). Right: the polarisation factor F of the two modes. Computed with radiosynth.

reads the temperature of the $s = 2$ or $s = 3$ shell, and as the frequency changes, the shell — and the height we read the temperature at — moves.

In one sentence: where a shell is opaque, we read the coronal temperature at the height of that shell.

6. Walking along the line of sight

What the computer does

At microwaves the plasma is far too thin to bend the rays, so radiosynth uses *straight* lines of sight, one per pixel. Along each line it:

1. **samples the model** at equally spaced points: density n_e, temperature T_e and the field $\boldsymbol{B}$ (strength and angle θ to the line of sight);
2. **looks between each pair of points for a shell**: a shell lies there if $s\nu_B - \nu$ changes sign. It finds where by interpolation, and estimates the thickness scale from the change of field over the step,
$$L_B = \frac{B_s\,\Delta\ell}{|B_{k+1} - B_k|}; \tag{5}$$
3. **computes the shell's optical depth** with Eq. (3), for both modes, plus the ordinary free–free absorption of the step (slightly different for the two modes in a field);
4. **adds the light** of each step, dimmed by everything in front of it,
$$I \;\leftarrow\; I + S\left(1 - e^{-\Delta\tau}\right)e^{-\tau}, \qquad \tau \leftarrow \tau + \Delta\tau, \tag{6}$$
once for each polarisation, from the observer inwards.

The shells are much thinner than a step, but that does not matter: the code does not need to see inside a shell, only to know that the line of sight crossed it.

In one sentence: straight lines; find the shells between sample points; add their opacity and glow, one polarisation at a time.

7. Left or right: the field tells us its direction

Circular polarisation

Where the field points towards the observer, the x-mode is right-handed; where it points away, left-handed. So radiosynth follows the right-handed (R) and the left-handed (L) light separately, each using the x-mode or the o-mode opacity depending on the field direction in each step, and gives

$$I = R + L, \qquad V = R - L, \qquad \frac{V}{I} \text{ between } -1 \text{ and } +1. \tag{7}$$

If a layer is opaque in the x-mode but transparent in the o-mode, almost all the light is in one hand and $|V/I| \to 1$; if it is opaque in both, $V \to 0$. The sign of V is the sign of the field. (Where the field turns across the line of sight, the light either keeps its hand — *weak coupling*, the default — or keeps its mode — *strong coupling*; both can be chosen.)

In one sentence: $V = R - L$ maps the polarity of the field; its size tells which modes are opaque.

8. The picture

A bipolar active region at 5 and 17 GHz

Fig. 2 shows what radiosynth computes for a model active region with two spots of opposite polarity, a 2.5 MK corona and fields up to 2500 G.

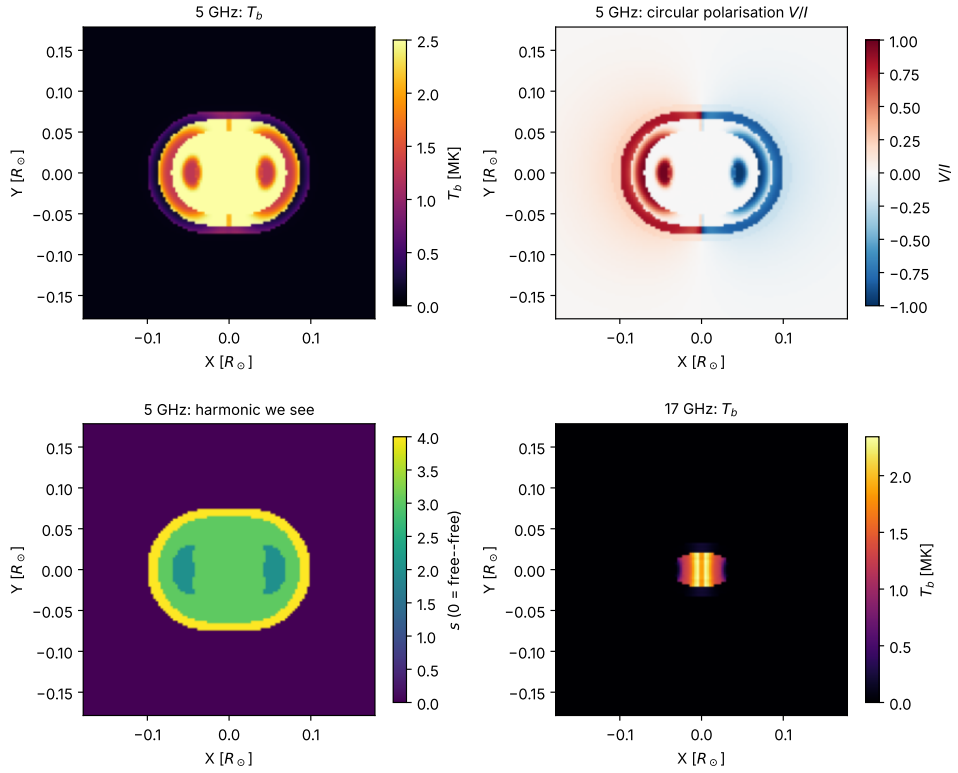


Figure 2: Gyroresonance and free–free emission of a model bipolar active region seen from above, computed with `rs.make_gr_image`. Top left: brightness temperature at 5 GHz; top right: circular polarisation V/I at 5 GHz; bottom left: the harmonic that produces most of the light at 5 GHz (0 means free–free); bottom right: brightness temperature at 17 GHz.

The story of the previous chapters can be read directly from the picture:

- **The source is as hot as the corona.** At 5 GHz most of the region shows $T_b = 2.5\,\mathrm{MK} = T_e$: the line of sight crosses the opaque $s = 3$ shell (green in the third panel). The total flux is 89 sfu, against 18 sfu and at most 0.45 MK from free–free alone.
- **The edge is the $s = 4$ shell** (yellow ring), which is only partly opaque, so the source fades there.
- **The spot centres are darker and fully polarised.** There the field points almost straight at us, $\sin\theta \to 0$: the shells become thin (Chapter 3), only the x-mode is still opaque, at $s = 2$, and the light is almost all of one hand, $|V/I| \approx 1$. Each spot is polarised with the sign of its field — red ($V > 0$) for the one pointing to us, blue for the other — and for this symmetric pair the total V is zero.
- **At 17 GHz the source shrinks.** The shells now need $B_3 \approx 2000\,\mathrm{G}$, which exists only low down, so only a small core between the spots, where the field is strong and seen across the line of sight, stays bright.

9. The whole story in six lines

1. Electrons spiral in the field at $\nu_B = 2.8\,\mathrm{MHz}$ per gauss.
2. At frequency ν they absorb and emit only in thin shells where $\nu = s\nu_B$.
3. The shell opacity falls by a factor of several hundred per harmonic, so only $s = 2$ and 3 are opaque; shells seen along the field are thin.
4. The x-mode, rotating with the electrons, is far more opaque than the o-mode.
5. An opaque shell shows the temperature of the corona at its height.
6. Following right- and left-handed light along straight lines gives I, and V, whose sign is the sign of the field.

In radiosynth all of this is one line:

```
img = rs.make_gr_image(grid, "5GHz, 17GHz", x_size=0.4, y_size=0.4)
img.plot("tb"); img.plot("pol"); img.plot("harmonic")
```

and every equation, with the code that implements it, is in the companion note *Thermal gyroresonance emission in radiosynth*.

Further reading. White & Kundu 1997, Sol. Phys. 174, 31 (gyroresonance emission from coronal magnetic fields); Dulk 1985, ARA&A 23, 169; Zheleznyakov 1970, *Radio Emission of the Sun and Planets*.